Smoothing Filters on Steroids

"Steroids are for guys who want to cheat their opponents," said Tom combatively.

The objective of smoothing filters in trading is to get the highest degree of smoothing possible within the constraint of inducing the least amount of lag. The purpose of this chapter is to develop those kinds of filters for traders. Then, it is up to the trader to interpret the results of the filtering. None of these filters are predictive. None of these filters explain market activity.

Nonrecursive Filters

Any nonrecursive filter having coefficients symmetrical about the center point of the filter and having a polynomial of odd degree will always have a root of the polynomial at the Nyquist frequency. This is important because completely rejecting the highest possible frequency component in the filter output goes a long way toward noise reduction. Noise reduction is the principal purpose of a smoothing filter.

Since the degree of the polynomial is one less than the number of elements in the filter, nonrecursive filters having the number of elements as 2, 4, 6, 8, 10, and so on will have this characteristic. Since the lag of nonrecursive filters is half the degree of the filter, these filters will have lags of 0.5, 1.5, 2.5, 3.5, and 4.5 bars, respectively. Filter design not only depends on the amount of smoothing desired, but also on the amount of lag that can be tolerated.

As a shorthand notation, $[b_0\, b_1\, b_2\, b_3 \ldots b_N] / S$ is introduced to describe the coefficients of nonrecursive filters. For example, [1 1 1 1] / 4 describes the coefficients of a four-element simple moving average (SMA). In fact, a four-element SMA fits the criterion of having a zero at the Nyquist frequency. Its frequency response is shown in Figure 3.1. This figure shows that a four-bar cycle period (frequency = 0.25) is also rejected. The critical frequency is the −3 dB point in the response. The −3 dB point along the frequency axis is where the output is attenuated to be half the input power. In this case the critical frequency is approximately 0.1 cycles per bar, corresponding to a 10-bar cycle period. The attenuation characteristic between a two-bar cycle period and a four-bar cycle period rises to about −12 decibels (dB), which means the wave amplitude at the output is about 25 percent of the amplitude at the input.

The expectation of obtaining more smoothing by increasing the length of the SMA leads us to examine a six-element nonrecursive filter whose elements are [1 1 1 1 1 1] / 6. The frequency response of this filter is shown in Figure 3.2. Increasing the degree of the polynomial introduced another zero in the transfer response polynomial so that two-, three-, and five-bar cycle periods are completely eliminated at the output. However, attenuation between the rejection points is not changed with respect to the lobe number, and is

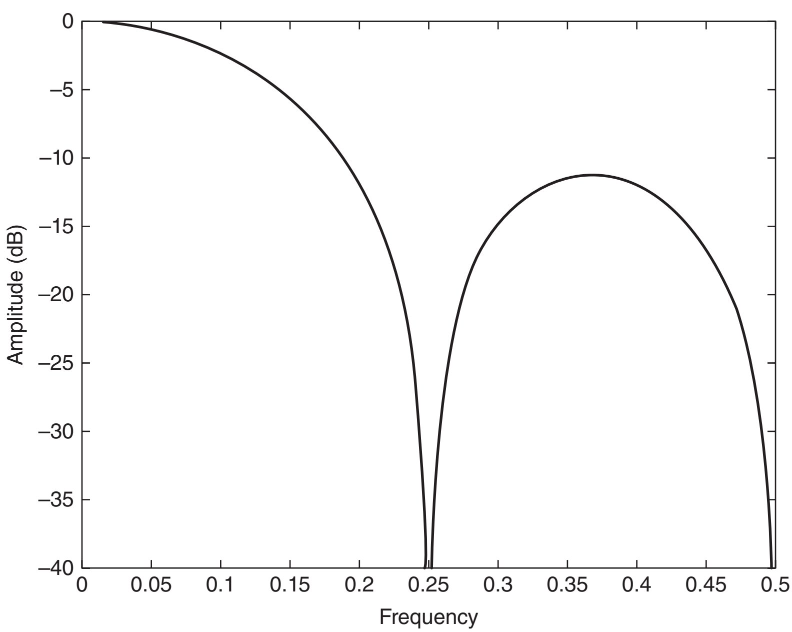

FIGURE 3.1 Frequency Response of a [1 1 1 1] / 4 Nonrecursive Filter

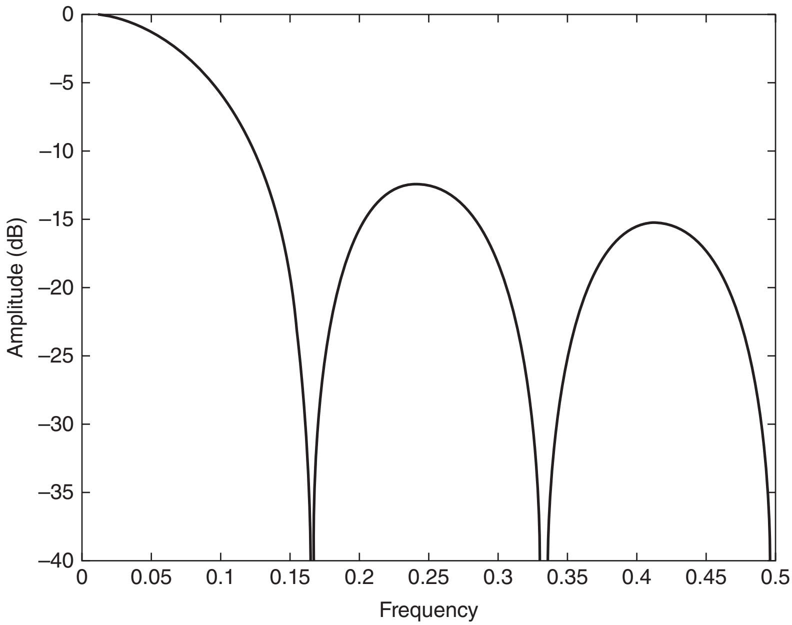

FIGURE 3.2 Frequency Response of a [1 1 1 1 1 1] / 6 Nonrecursive Filter

only slightly increased from lobe to lobe. The critical period of this filter is increased compared to the four-element SMA to be at a 13.7-bar cycle period.

Nonrecursive filters are not restricted to be SMAs. The coefficients can be weighted symmetrically about the center of the filter. Among the most simple of these is a filter whose coefficients are [1 2 2 1] / 6. The frequency response of this filter is shown in Figure 3.3. Comparing this nonrecursive filter to the SMA of like order (in Figure 3.1), we see that not only have the cycle periods that are completely rejected are two- and three-bar cycle periods as opposed to two- and four-cycle periods. In addition, the attenuation between the rejection frequencies rises to a minimum of −25 dB, meaning the output waveform amplitude is less than 5.6 percent of the input amplitude in this rejection band. The increased attenuation comes at the expense of a decrease in the critical cycle period to a 7.4-bar cycle. The filter delay remains the same at 2.5 bars.

From our experience with SMAs, we can improve both the smoothing and the rejection band attenuation by increasing the degree of the filter (at the expense of increased lag). The frequency response of a [1 2 3 3 2 1] / 12 nonrecursive filter is shown in Figure 3.4. In this case, two-, three-, and four-bar cycle periods are completely eliminated from the output of the

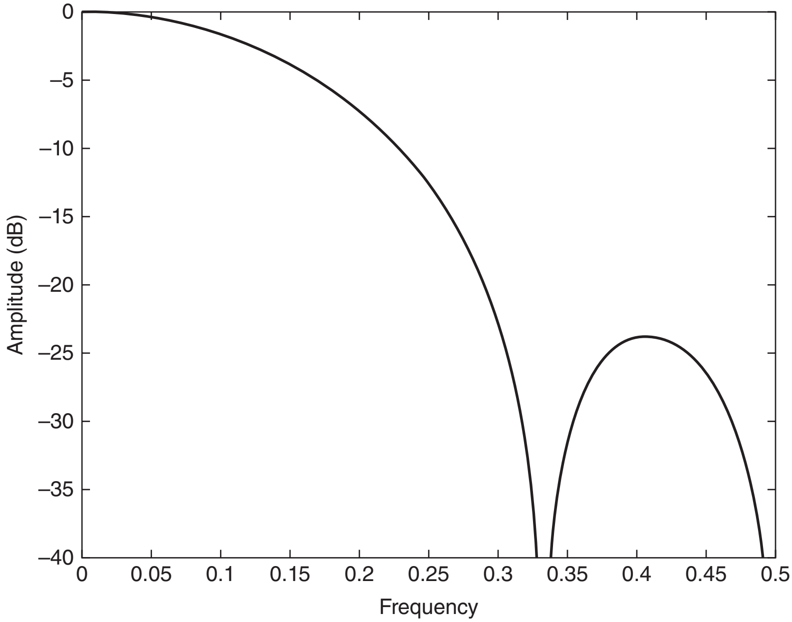

FIGURE 3.3 Frequency Response of a [1 2 2 1] / 6 Nonrecursive Filter

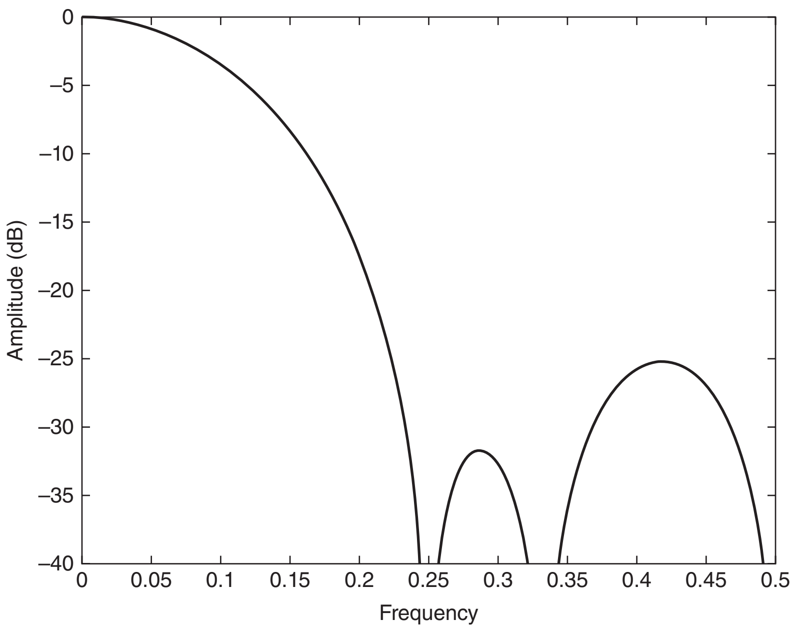

FIGURE 3.4 Frequency Response of a [1 2 3 3 2 1] / 12 Nonrecursive Filter

filter and the rejection in the stop bands is still greater than −25 dB. Filtering is improved because the critical period has been increased to be at a 10.9-bar cycle period. All this is done at the expense of a delay of 3.5 bars. While this is a pretty nice filter, we can do better.

Modified Simple Moving Averages

An interesting fact is that if the values of the first and last elements of an SMA having an even degree are cut in half, then one is guaranteed a double zero in the transfer response at the Nyquist frequency. For example, a five-element (four-degree) SMA would have coefficients as [0.5 1 1 1 0.5] / 4. The transfer response of a five-element modified SMA is shown in Figure 3.5 for comparison to the response of a four-element SMA shown in Figure 3.1. Not only do we have the double zero at the Nyquist frequency and the same zero at a four-bar cycle period, but we also have increased the minimum attenuation between the two zeros to be −20 dB, which means the output amplitude is only 10 percent of the input data amplitude. This improved

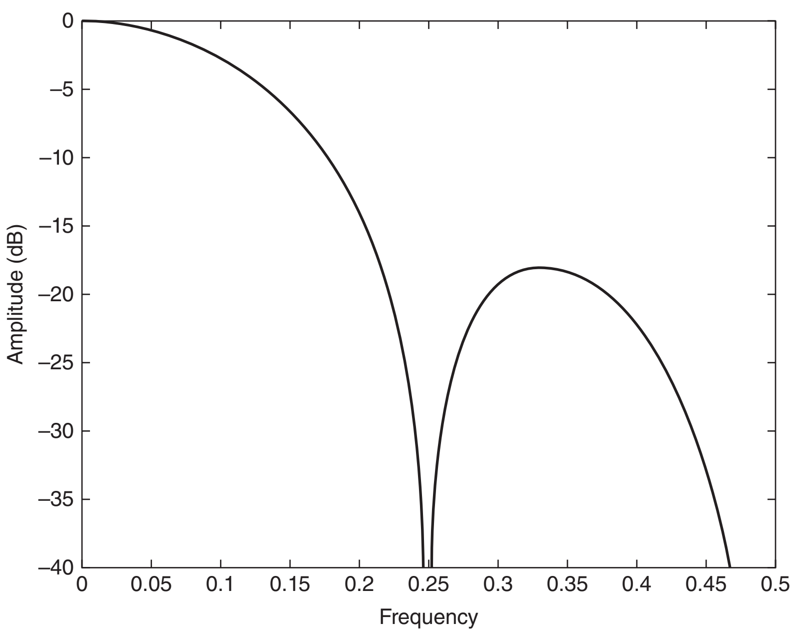

FIGURE 3.5 Frequency Response of a Five-Element Modified Simple Moving Average

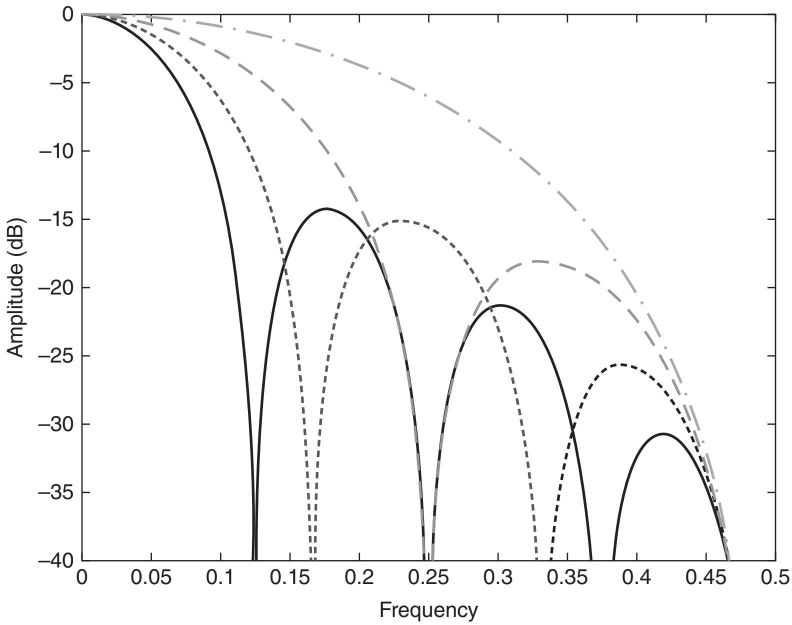

FIGURE 3.6 **Modified Simple Moving Average Frequency Responses**

filtering performance comes at the expense of only a half bar of delay because we went from a four-element filter to a five-element filter.

The frequency responses of three-, five-, seven-, and nine-element modified SMAs are shown in Figure 3.6.

Modified Least-Squares Quadratics

Instead of transfer response's having a least-squares best fit, as with the SMA, we can also arrange to obtain a least-squares best fit to a quadratic equation. This means we can accommodate some bowing in the data while getting our least-squares best fit. Then, we can halve the first and last coefficients to realize the double zero in the transfer response at the Nyquist frequency. Hamming[1] gives the coefficients of such filters as:

[7 24 34 24 7] / 96

[1 6 12 14 12 6 1] / 52

[−1 28 78 108 118 108 78 28 −1] / 544

[−11 18 88 138 168 178 168 138 88 18 −11] / 980

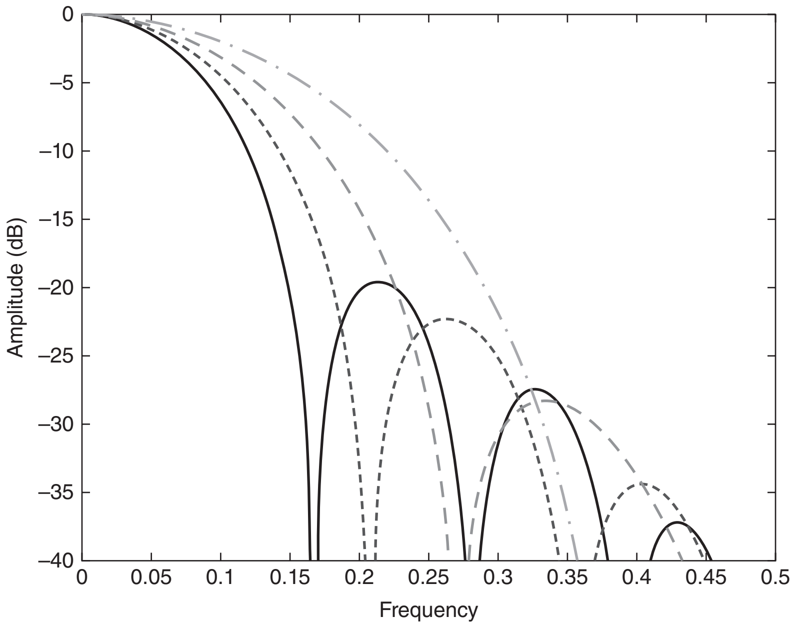

FIGURE 3.7 Modified Least-Squares Quadratics Filter Frequency Responses

The frequency responses of these filters are shown in Figure 3.7. The frequency responses in Figure 3.7 show that these are clearly superior smoothing filters for trading. These filters incur two, three, four, and five bars of lag, respectively.

SuperSmoother

One of the advantages of recursive filters is that the transition between the pass band and the stop band can be much sharper compared to that of the nonrecursive filters. As a matter of computational simplicity, the critical period of the filter can be established independently from the degree of the filter. On the downside, recursive filters don't have the same delay at all frequencies. This variable delay across the spectrum leads to dispersion distortion of the output waveform. Most of the dispersion occurs at those frequency components that have been attenuated, so the effect of the distortion is a relatively minor one.

A Butterworth filter is a favorite analog filter because its frequency response is maximally flat, near zero frequency. Years ago, I translated analog Butterworth filters to their digital approximations having like degrees in the numerator and denominator of the transfer response. The transfer response is characterized by a single variable: the critical period. The critical period is that where the output is attenuated by 3 dB relative to the input data.

The digital Butterworth filter approximations included numerators whose coefficients were [1 2 1] and [1 3 3 1] for two- and three-pole configurations, respectively. However, for applications to trading, I noted that the primary contribution of the numerator terms was to increase the filter lag. So I created modified versions simply by deleting all but the constant term in the numerator of the Butterworth transfer response.

The equations for the two-pole modified Butterworth filter are:

$$
\begin{aligned}
a &= e^{(-1.414*3.14159\ /\ Period)}\\
b &= 2 * a * Cos(1.414 * 1.25 * 180\ /\ Period)\\
c2 &= \mathrm{b}\\
c3 &= -a * a\\
c1 &= 1 - c2 - c3\\
Output &= c1 * Input - c2 * Output[1] - c3 * Output[2]
\end{aligned}
\tag{3-1}
$$

The equations for the three-pole modified Butterworth filter are:

$$
\begin{aligned}
a &= e^{(-3.14159\ /\ Period)}\\
b &= 2 * a * Cos(1.738 * 180\ /\ Period)\\
c &= a * a\\
d2 &= b + c\\
d3 &= -(c + b * c)\\
d4 &= c * c\\
d1 &= 1 - d2 - d3 - d4\\
Output &= d1 * Input - d2 * Output[1] - d3 * Output[2] - d4 * Output[3]
\end{aligned}
\tag{3-2}
$$

The frequency responses for two- and three-pole modified Butterworth filters whose critical period was set at a 10-bar cycle are shown in Figure 3.8. The three-pole version gives about 6 dB more attenuation at the Nyquist frequency than the two-pole version. That means the output wave amplitude of the three-pole version would be half the wave amplitude at the Nyquist frequency. In some applications, this difference in filtered amplitude could be important.

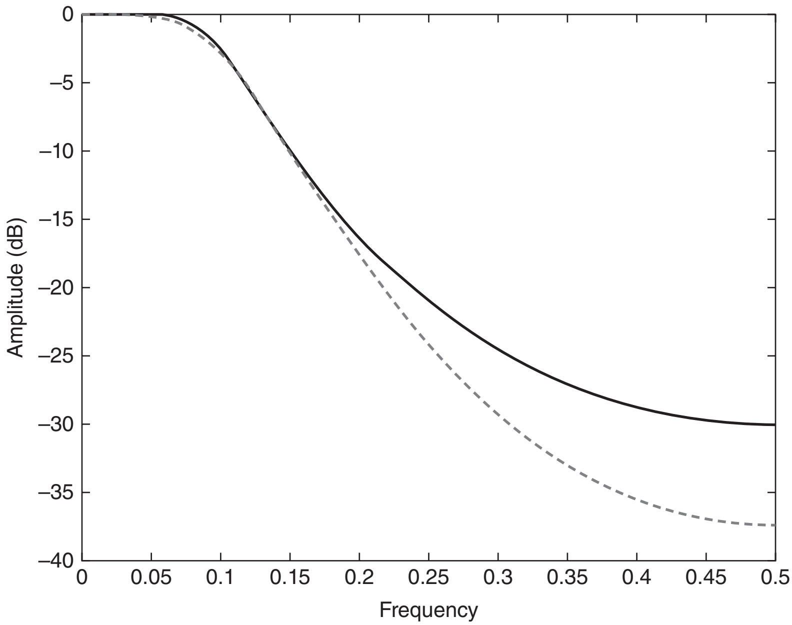

FIGURE 3.8 Frequency Response of Two- and Three-Pole Modified Butterworth Filters Where Period = 10

The delay of the two- and three-pole modified Butterworth filters whose critical period was set at a 10-bar cycle is shown in Figure 3.9. The price to be paid for the additional rejection band attenuation is an increase of almost two bars of additional delay near the critical period. In addition, there is greater percentage dispersion distortion across the pass band.

Having obtained good filtering with minimum delay and minimum delay distortion, it is relatively easy to obtain a zero in the transfer response at the Nyquist frequency by adding a two-element moving average in the numerator. The delay cost is only half a bar, and that cost seems worth it. By doing this, a filter I call the SuperSmoother is created. The equations for the SuperSmoother filter are:

$$\begin{aligned}
a &= e^{(-1.414*3.14159\ /\ Period)} \\
b &= 2 * a * Cos(1.414 * 180\ /\ Period) \\
c2 &= b \qquad\qquad (3\text{-}3) \\
c3 &= -a * a \\
c1 &= 1 - c2 - c3 \\
Output &= c1 * (Input + Input[1])\ /\ 2 + c2 * Output[1] + c3 * Output[2]
\end{aligned}$$

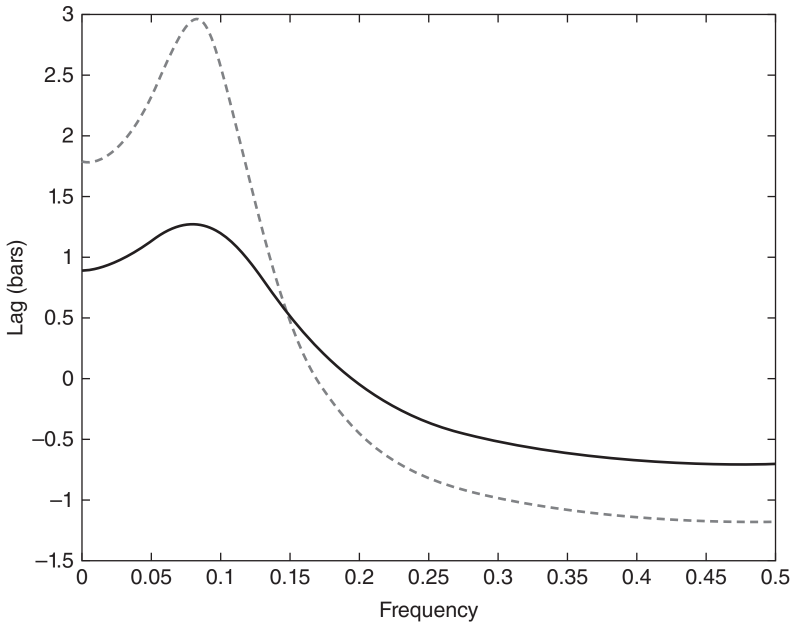

FIGURE 3.9 Lags of Two- and Three-Pole Modified Butterworth Filters Where Period = 10

The frequency response of the SuperSmoother filter where the critical period is set to 10 is shown in Figure 3.10.

SuperSmoother Filter Applications

A SuperSmoother filter is used anytime a moving average of any type would otherwise be used, with the result that the SuperSmoother filter output would have substantially less lag for an equivalent amount of smoothing produced by the moving average. For example, a five-bar SMA has a cutoff period of approximately 10 bars and has two bars of lag. A SuperSmoother filter with a cutoff period of 10 bars has a lag a half bar larger than the two-pole modified Butterworth filter shown in Figure 3.9. Therefore, such a SuperSmoother filter has a maximum lag of approximately 1.5 bars and even less lag into the attenuation band of the filter. The differential in lag between moving average and SuperSmoother filter outputs becomes even larger when the cutoff periods are larger.

Market data contain noise, and removal of noise is the reason for using smoothing filters. In fact, market data contain several kinds of noise. I'll

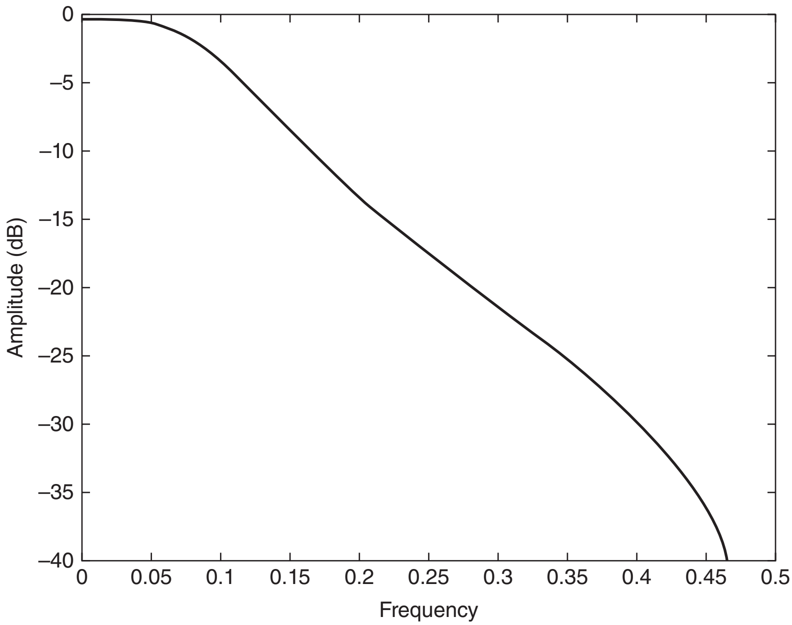

FIGURE 3.10 Frequency Response of a SuperSmoother Filter Where Period = 10

group one kind of noise as systemic, caused by the random events of trades being exercised. A second kind of noise is aliasing noise, caused by the use of sampled data. Aliasing noise is the dominant term in the data for shorter cycle periods.

It is easy to think of market data as being a continuous waveform, but it is not. Using the closing price as representative for that bar constitutes one sample point. It doesn't matter if you are using an average of the high and low instead of the close, you are still getting one sample per bar. Since sampled data is being used, there are some DSP aspects that must be considered. For example, the shortest analysis period that is possible (without aliasing)[2] is a two-bar cycle. This is called the Nyquist frequency, 0.5 cycles per sample. A perfect two-bar sine wave cycle sampled at the peaks becomes a square wave due to sampling. However, sampling at the cycle peaks cannot be guaranteed, and the interference between the sampling frequency and the data frequency creates the aliasing noise. The noise is reduced as the data period is longer. For example, a four-bar cycle means there are four samples per cycle. Because there are more samples, the sampled data are a better replica of the sine wave component. The replica is better yet for an eight-bar data component. The improved fidelity of the sampled data means the aliasing noise is reduced at longer and longer cycle periods. The rate of reduction is 6 dB per octave. My experience is that the systemic noise rarely is more than 10 dB below the level of cyclic information, so that we create two conditions for effective smoothing of aliasing noise:

1. It is difficult to use cycle periods shorter that two octaves below the Nyquist frequency. That is, an eight-bar cycle component has a quantization noise level 12 dB below the noise level at the Nyquist frequency. Longer cycle components therefore have a systemic noise level that exceeds the aliasing noise level.
2. A smoothing filter should have sufficient selectivity to reduce aliasing noise below the systemic noise level. Since aliasing noise increases at the rate of 6 dB per octave above a selected filter cutoff frequency and since the SuperSmoother attenuation rate is 12 dB per octave, the SuperSmoother filter is an effective tool to virtually eliminate aliasing noise in the output signal.

I urge readers to universally adapt the SuperSmoother filter set to a cutoff period of 10 bars or so on all data to attenuate aliasing noise. The output of the SuperSmoother filter can be used directly as an indicator or as the sampled data fed to any other indicator.

Key Points to Remember

1. The lag of all nonrecursive filters having coefficients symmetrical about the center point of the filter is half the degree of the transfer response polynomial. In other words, for a nonrecursive filter having N elements, the lag of that filter will be $(N - 1) / 2$.
2. Nonrecursive filters with coefficients symmetrical about the center point of the filter and having an even number of elements will always have a zero in the transfer response at the Nyquist frequency.
3. An SMA having an odd number of elements and whose first and last elements have coefficients cut in half are guaranteed of having a double zero in the transfer response at the Nyquist frequency.
4. Whereas SMAs are a least-squares best fit to a straight line, nonrecursive filters can also have coefficients that are a least-squares best fit to a quartic. Halving their end coefficients also guarantees a double zero in their transfer response at the Nyquist frequency.

5. Probably the best filter for most smoothing applications is a two-pole SuperSmoother.
6. A SuperSmoother filter is recommended for universal use to remove aliasing noise.

Notes

1. R. W. Hamming, *Digital Filters,* 3rd ed. (Upper Saddle River, NJ: Prentice Hall, 1997), 49.
2. Aliasing is the false signals that result from undersampling. For example, wagon wheels often appeared to turn backwards in the early cowboy motion pictures because of the relatively slow movie frame rate.